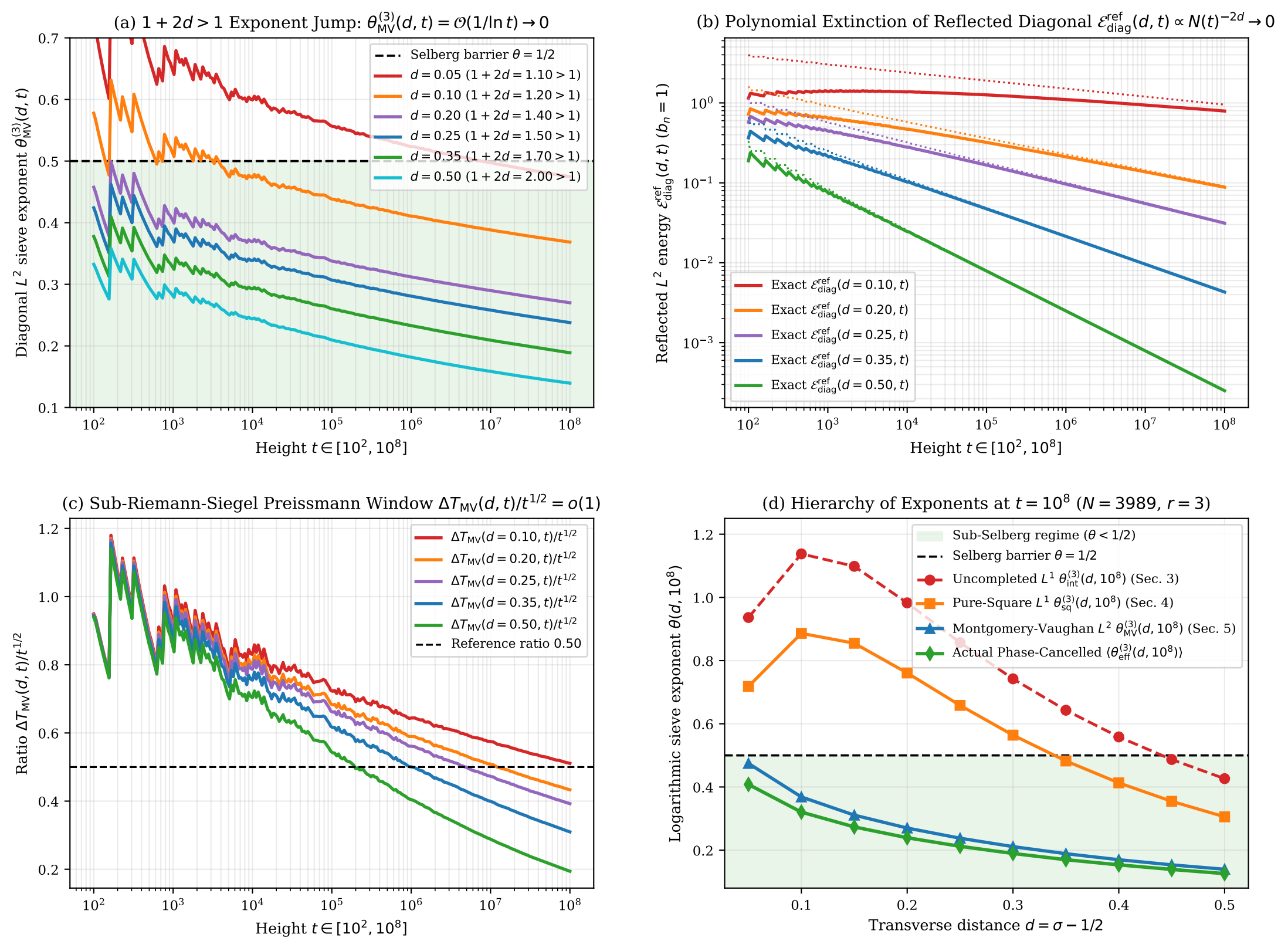

(a) $1+2d>1$ Exponent Jump: $\theta^{(3)}_{\mathrm{MV}}(d,t)=\mathcal{O}(1/\ln t)\to 0$
Diagonal L^2 sieve exponent $\theta^{(3)}_{\mathrm{MV}}(d,t)$
Height $t\in[10^2,10^8]$
Selberg barrier $\theta=1/2$
$d=0.05$ $(1+2d=1.10>1)$
$d=0.10$ $(1+2d=1.20>1)$
$d=0.20$ $(1+2d=1.40>1)$
$d=0.25$ $(1+2d=1.50>1)$
$d=0.35$ $(1+2d=1.70>1)$
$d=0.50$ $(1+2d=2.00>1)$
(b) Polynomial Extinction of Reflected Diagonal $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t)\propto N(t)^{-2d}\to 0$
Reflected L^2 energy $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d,t)$ $(b_n=1)$
Height $t\in[10^2,10^8]$
Exact $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d=0.10,t)$
Exact $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d=0.20,t)$
Exact $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d=0.25,t)$
Exact $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d=0.35,t)$
Exact $\mathcal{E}^{\mathrm{ref}}_{\mathrm{diag}}(d=0.50,t)$
(c) Sub-Riemann-Siegel Preissmann Window $\Delta T_{\mathrm{MV}}(d,t)/t^{1/2}=o(1)$
Ratio $\Delta T_{\mathrm{MV}}(d,t)/t^{1/2}$
Height $t\in[10^2,10^8]$
$\Delta T_{\mathrm{MV}}(d=0.10,t)/t^{1/2}$
$\Delta T_{\mathrm{MV}}(d=0.20,t)/t^{1/2}$
$\Delta T_{\mathrm{MV}}(d=0.25,t)/t^{1/2}$
$\Delta T_{\mathrm{MV}}(d=0.35,t)/t^{1/2}$
$\Delta T_{\mathrm{MV}}(d=0.50,t)/t^{1/2}$
Reference ratio 0.50
(d) Hierarchy of Exponents at $t=10^8$ $(N=3989,\ r=3)$
Logarithmic sieve exponent $\theta(d,10^8)$
Transverse distance $d=\sigma-1/2$
Sub-Selberg regime $(\theta<1/2)$
Selberg barrier $\theta=1/2$
Uncompleted L^1 $\theta^{(3)}_{\mathrm{int}}(d,10^8)$ (Sec. 3)
Pure-Square L^1 $\theta^{(3)}_{\mathrm{sq}}(d,10^8)$ (Sec. 4)
Montgomery-Vaughan L^2 $\theta^{(3)}_{\mathrm{MV}}(d,10^8)$ (Sec. 5)
Actual Phase-Cancelled $\langle\theta^{(3)}_{\mathrm{eff}}(d,10^8)\rangle$